Response to CMS RFI: Health Technology Ecosystem (CMS-0042-NC)

SUMMARY

I. Introduction

Massive Bio, Inc. is an AI-driven oncology clinical trial and real-world data technology company committed to democratizing access to advanced cancer care and research. Through our SYNERGY-AI platform, we engage over 160,000 consented oncology patients across 17 countries and integrate clinical, genomic, and real-world data from over 3,400 sites and 100,000 physicians. Recognized by the White House Cancer Moonshot, our mission is to transform access to precision oncology through patient-centric, digitally enabled platforms.

We appreciate the opportunity to provide public comment on CMS's Request for Information 2025-08701 regarding the Health Technology Ecosystem. We commend CMS, ASTP, and ONC for taking steps to empower patients, foster interoperability, and create a trusted, seamless digital health landscape for Medicare beneficiaries.

Our comments focus primarily on:

- **PC-1 through PC-12 (Patients and Caregivers)**
- **PR-1 through PR-4 (Providers)**
- **TD-1 through TD-10 (Technology Vendors and Data Providers)**

II. Key Recommendations and Responses

PC-1 / PC-2 / PC-3 / PC-5 / PC-6 – AI-driven Navigation & Real-Time Trial Matching

Massive Bio's Patient Connect app helps Medicare beneficiaries understand their condition, organize medical records, and match in real time to clinical trials based on genomic, imaging, and EHR data. This hybrid "click-and-mortar" system combines digital screening with concierge support, enabling participation from rural or underserved areas. Patients have reported easier decision-making and faster access to therapies. We recommend CMS promote such tools by creating a "CMS Trusted App" designation and integrating trial awareness into the Annual Wellness Visit.

PC-4 / PC-8 / PC-9 / PC-10 / PC-12 – Integrated Data, TEFCA, and Real-Time Use Cases

Structured access to full EHR data, especially pathology reports and genomics, remains elusive. By leveraging TEFCA and FHIR-based interoperability, Massive Bio aggregates records from multisite networks and identifies high-probability matches for trials. We support extending Blue Button 2.0 to include MA encounter data, genomics, social determinants, and lab results, and enabling patient-mediated FHIR access to HIEs.

PR-1 / PR-2 / PR-4 – Reducing Provider Burden with Smart Integration

Providers in our network benefit from AI tools that auto-extract eligibility from EHRs and reduce manual chart review. CMS should further support digital product integration into physician workflows through CPT codes for digital navigation and via funding implementation grants (especially in rural ACOs). Interoperability policies should mandate APIs that support

both structured and unstructured data.

TD-1 / TD-2 / TD-3 / TD-4 / TD-9 / TD-10 – Unlocking Innovation for Developers
Developers require clearer access pathways to CMS data and APIs beyond claims (e.g., imaging reports, genomic annotations, social care data). CMS could establish a Developer Sandbox with synthetic data and credentialed digital identity frameworks (e.g., Login.gov + SMART on FHIR) to facilitate application development. We encourage support for OpenID Connect, and expansion of the ONC Certification Program to prioritize real-time patient-use APIs.

VB-1 / VB-3 / VB-5 / VB-11 – Aligning Digital Tools with Value-Based Oncology
Massive Bio supports inclusion of digital oncology tools into ACO models and the Enhancing Oncology Model (EOM). CMS could incentivize trial referrals, tele-consent, and AI-driven remote monitoring as measures of innovation and equity in care navigation. Pilot projects could link Medicare claims to real-time eligibility APIs to inform shared savings calculations.

III. Conclusion

We commend CMS and ONC for their leadership in advancing the patient-centric digital ecosystem. To unlock the full potential of health technology for Medicare beneficiaries, CMS should:

- **Prioritize use of certified AI-powered tools that enable decentralized clinical trial access.**
- **Expand patient access to all EHR formats under TEFCA and Blue Button.**
- **Invest in caregiver support features and multi-language accessibility.**
- **Promote public-private partnerships that drive innovation, such as ACTR and trial-matching.**
- **Massive Bio is eager to continue supporting these efforts. We are available for demonstration or pilot engagement opportunities with CMS and ONC.**

Patients and Caregivers (PC)
Patient Needs and Digital Health Apps: Medicare beneficiaries and their caregivers need intuitive digital tools for managing health, yet many lack access to comprehensive, user-friendly applications. Respondents to this RFI highlighted a desire for health management and care navigation apps that help understand conditions, coordinate care, and prompt next steps. Today's apps often fall short on features and interoperability; for example, patients struggle to gather all their health information in one place and to know what actions to take. We recommend expanding support and incentives for development of patient-centered apps tailored to older adults and caregivers. These apps should integrate data from multiple providers, present clear care plans, and use plain language explanations. CMS could partner with industry (e.g. via challenge grants or the CancerX public-private partnership) to spur innovation. Massive Bio's AskFiona AI is an example of an empathetic patient-facing chatbot that guides cancer patients through clinical trial options, clarifying trial purposes and eligibility in simple terms. Such AI-driven assistants can demystify complex healthcare decisions, empower patients with knowledge, and personalize recommendations at scale. To encourage adoption, CMS should promote digital

literacy and training for beneficiaries and caregivers (e.g. through Medicare's education channels or partnerships with community organizations) and ensure these tools are accessible (mobile-friendly, multilingual, and accommodating disabilities). Importantly, building trust in digital health products is critical – CMS could establish a voluntary approval or endorsement framework (as alluded in question PC-5) for third-party apps that meet usability, accessibility, and privacy standards, giving patients confidence in using them.

Data Access and Integration: A seamless, patient-centric digital ecosystem requires that patients and caregivers can easily access all of their health data in one place, without managing dozens of portals. Despite policy advances (HIPAA's right of access, 21st Century Cures Act, and CMS Interoperability rules), the reality is that many patients still face fragmented records and cumbersome processes. CMS's Blue Button 2.0 API made Medicare claims and coverage data available via FHIR®, spawning apps that let beneficiaries view claims history and out-of-pocket spending. However, Blue Button 2.0 currently offers only basic demographics, coverage, and Part A/B/D claims data. We recommend expanding Blue Button 2.0 to include additional data valuable for care management – for example, incorporating chronic condition indicators, lab results from claims (e.g. test codes and values when available), prior authorization or DME utilization data, and linking immunization or screening records. Developers have indicated that richer datasets would enable more useful patient-facing tools. CMS should also explore integrating Medicare Advantage and Medicaid data into a unified patient access API, so beneficiaries who transition between programs or have multiple coverage types see their information in one place. To handle the influx of data, CMS can leverage the United States Core Data for Interoperability (USCDI) standard – a common set of data elements for nationwide exchange – ensuring that new data fields are standardized and developer-friendly.

Another pillar of integration is harnessing Health Information Exchanges (HIEs) and TEFCA for patient access. The Trusted Exchange Framework and Common Agreement (TEFCA) has launched a nationwide network-of-networks to enable sharing of clinical data across disparate HIEs. Early progress is promising – as of 2024, thousands of organizations and over 14 million documents have been exchanged via TEFCA's initial Qualified Health Information Networks. TEFCA now even supports Individual Access Services, meaning patients (or apps on their behalf) can retrieve health records through this network, not just providers. CMS should promote patient-facing uses of TEFCA by coordinating with ONC to certify or endorse apps that serve as patient gateways to TEFCA data. For example, an approved app could, with one login, pull a patient's records from all TEFCA-connected providers – offering a longitudinal health record. This would fulfill the vision of a one-stop patient portal. HIEs can also play a role: many regional HIEs already aggregate data from hospitals, labs, and clinics; CMS could encourage HIEs to offer consumer access solutions or integrate with Medicare's Blue Button data. To advance this, CMS might require that HIEs participating in Medicare value-based programs enable patient access as a condition of participation, or provide grant funding to HIEs developing patient/caregiver portals. Respondents noted that data from HIEs, when available, is valuable but can vary in accuracy and usability. CMS could work with ONC on data quality standards for HIE-shared data (e.g. reconciling duplicate records, ensuring timely updates) to boost trust in these sources.

Digital Identity and Privacy: Patients and caregivers often juggle multiple logins across portals and

apps, a major barrier to adoption of digital tools. A single, secure digital identity that is widely accepted in healthcare could simplify access while safeguarding privacy. The RFI inquires about using identity credential services (like CLEAR, Login.gov, ID.me) that comply with NIST 800-63-3 Identity Assurance Level 2 (IAL2) – a high assurance of identity verification. We support encouraging (but not forcing) patients to obtain a secure digital identity credential which could be used to access any healthcare portal or app. For example, CMS could integrate with Login.gov (a federal OpenID Connect service) to let Medicare beneficiaries log into Medicare.gov, Blue Button apps, and even third-party health apps using that one verified account. This approach would reduce the proliferation of usernames/passwords and improve security via strong, multi-factor authentication. The benefits to patients and caregivers would be convenience and confidence that their identity is protected. We acknowledge potential downsides: some seniors may find credentialing processes (especially remote identity proofing) confusing, and there may be privacy concerns about a single ID being used widely. To address this, CMS should provide clear education on how digital identity works and ensure opt-in use. We also recommend involving caregivers in this framework – for instance, allowing an IAL2-credentialed caregiver proxy to access an elderly patient's data with consent.

Hand in hand with identity is privacy and information blocking enforcement. CMS and ONC have made it clear that patients should be able to access their electronic health information without interference. Yet, awareness of how to report information blocking is low among the public. CMS should conduct outreach (through Medicare handbooks, patient advocates, etc.) to encourage patients and caregivers to report information blocking when they encounter providers or IT developers improperly restricting access. A simple, well-publicized online complaint portal (building on ONC's existing portal) and protections against retaliation would empower consumers to speak up. More reports, if investigated by ONC, could spotlight problem areas and ultimately deter information blocking practices. Finally, to protect patient privacy while enabling data use, CMS should promote adoption of consent standards (like the HL7 FHIR Consent resource) for apps and networks. Patients should have granular control over how their data is shared through APIs or frameworks like TEFCA. Clear consent workflows – for example, an app asking a patient to approve sharing of specific data types with a research study – must be standardized and made easily understandable. Massive Bio's platform demonstrates a patient-centered consent approach: by orchestrating remote eConsent in its "just-in-time" trial enrollment model, patients anywhere can review trial details and electronically consent to participate, accelerating research while maintaining informed choice. CMS can draw on such best practices to inform future guidance on e-consent in healthcare and research.

<u>Recommendations for Patients/Caregivers:</u>

- Promote App Development and Literacy: Sponsor innovation challenges and community training to increase the availability of senior-friendly health management apps. Endorse apps that meet high standards to build trust.
- Unify Data Access: Expand Blue Button 2.0 to include more data (e.g. labs, Part C, Medicaid) and integrate with HIE/TEFCA networks so patients can retrieve a one-stop health record.
- Streamline Digital Identity: Offer a single Medicare digital identity (IAL2-verified) via OpenID Connect for use across portals, and engage caregivers in identity and access

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management.

- Empower Patients in Interoperability: Publicize the process to report information blocking, and enforce rules that providers must grant third-party apps access to EHR data per patient requests (per 21st Century Cures Act and CMS conditions).
- Enhance Consent and Privacy: Standardize eConsent for data sharing and ensure apps clearly convey what data is used, giving patients control. Emphasize privacy protections in any identity or data-sharing initiative to maintain trust.

Providers (PR)

Encouraging Digital Health Adoption by Providers: Frontline healthcare providers are critical to integrating digital health solutions into patient care, yet many remain cautious. CMS can play a pivotal role in incentivizing providers to leverage approved digital products in clinical practice. For example, CMS might incorporate use of certified care management apps or remote monitoring tools into value-based payment models or MIPS improvement activities. In rural areas and smaller practices, lack of resources and awareness are key obstacles. We recommend that CMS expand technical assistance and grant programs to support adoption of digital health in underserved settings – similar to past ONC extension centers for EHRs. This could include template workflows for using patient apps (e.g. how to refer a patient to a vetted diabetes management app and monitor progress) and cybersecurity best practices for third-party integrations. Providers also need clarity on their responsibilities when recommending digital tools. CMS, in partnership with medical boards, should issue guidance delineating providers' liability and oversight duties: for instance, if a clinician suggests an app, they should inform patients about data privacy and disclaim that the app is not a medical device unless FDA-approved. Providers should at minimum ensure any recommended app has appropriate evidence or certification (perhaps tied to a CMS-approved list as noted above). When providers do use digital tools in care, they should share relevant information with patients – e.g. explaining how an AI-driven decision support reached a recommendation. This transparency is vital to maintain trust in AI in healthcare.

Streamlining Clinical Workflows and Reducing Burden: One of the biggest barriers providers face is the burden of data entry and quality reporting, which diverts time from patient care. Digital health solutions must reduce, not add to, provider workload. We urge CMS to focus policy on integration of apps into EHR workflows. For instance, apps that require separate logins or duplicate data entry discourage provider use. The adoption of standards like SMART on FHIR is key here. SMART on FHIR allows third-party apps to launch within the EHR interface and access data via secure FHIR APIs, using OAuth2 and OpenID Connect for authentication (pmc.ncbi.nlm.nih.gov). CMS should continue to enforce the 21st Century Cures Act requirement that certified EHRs support SMART on FHIR (the "standardized API" criterion). Additionally, CMS could require EHR vendors to simplify the app onboarding process. Today, many hospitals face lengthy procedures to enable an app in their EHR, even if the app meets technical standards. Through conditions of participation or certification incentives, CMS/ONC could push for a more turnkey "app store" model for healthcare – where providers can plug in approved apps with minimal IT hassle.

A related challenge is accessing all relevant data for care. Providers report difficulty obtaining outside records or even within an EHR if data is locked in PDFs or images. We echo the need

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for full EHR data accessibility – including scanned documents, diagnostic images, and free-text notes – through interoperable means. Clinicians shouldn't have to manually hunt through faxes or portals. CMS can advance this by supporting initiatives to standardize unstructured data exchange. For example, requiring that certified health IT export documents in both human-readable and structured formats when possible (C-CDA or emerging FHIR DocumentReference with embedded text). We also encourage CMS to work with ONC on R&D for natural language processing (NLP) tools that convert free text (like consult notes) into queryable data. Massive Bio's SYNERGY-AI platform demonstrates the power of NLP and AI in structuring documents and extracting patient data for matching to trials (businesswire.com). In a hospital setting, similar AI could help parse clinical notes and populate key fields for care coordination. While making all data interoperable has cost and privacy implications (scanned records may contain sensitive or extraneous info), the priority should be on critical data for treatment (recent labs, meds, problem lists). We suggest CMS pilot projects to identify the most valuable data currently "trapped" in unstructured formats and fund solutions to liberate it safely.

Interoperability and Data Exchange: Providers are increasingly expected to exchange data with other providers, patients, and payers. CMS has promoted FHIR API use cases in its regulations – e.g., patient access APIs, provider directory APIs, payer-to-payer exchange for care continuity. We encourage providers to adopt these and more. In fact, many leading health systems already support multiple FHIR APIs (for individual and bulk data) either directly or via intermediaries. However, smaller practices may not be aware or have not tapped into these capabilities. CMS should require EHR vendors (via ONC certification) to provide implementation toolkits to their provider clients for each of these standard APIs, explaining use cases like automating patient intake forms or enabling population health queries. Moreover, bulk FHIR data export is a game-changer for reducing manual reporting burden. We strongly support CMS's exploration of using Bulk FHIR for quality reporting. Instead of each provider calculating quality measures and submitting through cumbersome processes, providers could simply export raw patient data (de-identified or as allowed) and let CMS perform the analytics centrally. Early tests of this approach are promising – an AI-based platform by ConcertAI showed that automating data retrieval from EHR and into research systems cut staff time significantly, reducing screening time per patient from ~41 minutes to ~12 minutes with no loss of accuracy. By relieving providers of double-documentation and manual uploads, such approaches let clinicians focus on care. We recommend CMS pilot a voluntary program where providers allow CMS to pull quality data via FHIR APIs (with proper privacy safeguards) and compare the effort and accuracy against current reporting. This could demonstrate savings in clinician hours and eventually be scaled to all of Medicare.

Another way to assist providers is by aligning interoperability efforts with their existing quality workflows. For example, many providers already participate in registries or use electronic case reporting for public health. CMS could design interoperability requirements so that one action serves multiple purposes. If a provider sends standardized data to a clinical registry for patient care feedback, perhaps that same feed could satisfy a CMS quality measure submission, killing two birds with one stone. The key is to consolidate requirements and reuse data flows. We propose CMS convene a workgroup with providers, EHR developers, and measure stewards to identify overlaps between interoperability tasks and quality reporting tasks, aiming to simplify the total ask on providers.

Digital Identity for Providers and Partners: Just as with patients, having a unified digital identity for providers can streamline interoperability. Many providers now must manage certificates or logins for various HIE networks, payer portals, etc. If CMS championed a nationwide provider digital identity credential (meeting IAL2/AAL2 standards), providers could authenticate more easily across systems. This might tie into plans for a national provider directory that includes digital contact points (FHIR endpoints) and verified identities. We support creation of a nationwide FHIR endpoint & provider identity directory, potentially built by a public-private collaboration, to serve as a phonebook for interoperability. Such a directory, if it included each provider's organization affiliations, FHIR server URLs, and trusted identity assertions, would help both human and software workflows. For example, when Dr. Smith refers a patient to Dr. Jones, the directory could be used to seamlessly share records by knowing Jones's endpoint and verifying identity. It could also assist in prior authorization and payer-provider data exchange, as payers could quickly identify treating providers and share data if the directory links identities to endpoints. We note that requiring use of digital identity credentials and OIDC protocols could raise some concerns among providers – e.g., workflow changes or initial onboarding burden. Therefore, CMS might phase this in by first enabling voluntary use (perhaps an incentive if providers use federated login for CMS systems), then gradually requiring it for certain transactions (like access to a national provider portal or directory).

Combatting Information Blocking and Enhancing Compliance: Despite the information blocking regulations in effect, enforcement is uneven. CMS can complement ONC's efforts by establishing disincentives for providers who engage in information blocking. Currently, certain provider types (like some physician groups) do not face clear financial penalties for blocking behavior. CMS could leverage its payment programs: for example, making compliance with information blocking a condition of Medicare participation or tying it to MIPS scoring. An effective approach might be a graduated penalty – e.g., warnings, then public reporting of non-compliance, then a payment reduction for repeat offenders who refuse to share data. We also propose CMS gather stakeholder input on whether any exceptions in the ONC information blocking rules should be revised. Providers have voiced that some exceptions (like the "Infeasible" exception) are vague and could be misused. Tightening these could further discourage unjustified refusal to share data. Finally, encouraging providers to report information blocking by others (such as a hospital reporting an EHR vendor's restrictive practices, or a physician reporting a lab that won't transmit results digitally) can help identify problem areas. CMS can amplify ONC's reporting portal among providers and assure whistleblower-type protections.

<u>Recommendations for Providers:</u>

- Incentivize Digital Tool Use: Reward providers in payment models for using validated digital health apps and remote monitoring with patients. Fund training for small/rural practices to adopt telehealth, AI decision support, etc.
- Embed Apps into Workflow: Enforce and enhance SMART on FHIR API capabilities so third-party apps integrate smoothly in EHRs. Push for easier app deployment (app marketplaces) so providers can adopt tools without high IT overhead.
- Reduce Data Entry Burden: Expand use of Bulk FHIR to let providers send raw data to CMS for quality reporting, instead of manual chart abstraction. Re-use existing clinical data

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streams (registries, HIE) for multiple reporting needs to avoid duplicative work.

- Make All Data Usable: Develop standards for sharing traditionally non-interoperable data (scans, notes). Support AI/NLP to extract key info from unstructured data in EHRs, improving completeness of data exchange.
- Unified Provider Identity: Implement a federated provider credential for logging into CMS and HIE systems, and build a national provider & FHIR endpoint directory. This will streamline referrals, data queries, and verify actors in exchanges.
- Strengthen Data Sharing Compliance: Establish Medicare penalties for willful information blocking by providers. Publicize non-compliance if needed. Also, simplify the process for providers to report blocking by IT vendors or others, feeding into ONC enforcement.

Payers (PA)

Advancing Interoperability in Payer Systems: Health plans (including Medicare Advantage organizations and Medicaid plans) hold vast data that could improve care continuity if shared appropriately. CMS has led by example with initiatives like Blue Button 2.0 for claims and the Data at the Point of Care API (which allowed providers to retrieve Medicare FFS claims for their patients). We recommend CMS continue to push payers to implement and utilize open APIs for patient and provider access. Specifically, CMS should expand regulatory requirements (from the CMS Interoperability and Patient Access final rule) to all Medicare Advantage and Qualified Health Plans to not only maintain a Patient Access API but also a Provider Access API and a Payer-to-Payer exchange API for seamless data exchange. Payers should make claims, encounter, and formulary data available in real time to members and treating providers. CMS can encourage this by highlighting success stories: for instance, payers that have enabled real-time data sharing see better care coordination and fewer duplicated tests. Moreover, CMS might consider hosting a sandbox or gateway that simplifies how providers query multiple payers' data – essentially a "universal payer API" endpoint that routes requests to the right plan, potentially via TEFCA or a centralized directory of payer FHIR endpoints.

Evaluating TEFCA's Role for Payers: The RFI asks what policy or technical limitations exist in TEFCA for payers and how to address them. One limitation noted by stakeholders is that TEFCA initially focuses on provider-to-provider exchange; payers joining as Qualified Health Information Networks (QHINs) or participants may face unclear use cases or concerns about data overlap with existing networks. Payers worry about the value proposition of TEFCA participation relative to costs. We suggest CMS work with ONC to tailor TEFCA to support payer-specific needs such as risk adjustment data acquisition, prior authorization data sharing, and quality reporting. For example, TEFCA could facilitate retrieval of clinical records for HEDIS measures or for reviewing prior authorization requests, which would be a tangible benefit to plans. If those use cases are standardized within TEFCA, payer participation will be more attractive. To address technical limitations, CMS could sponsor pilot projects where a Medicare Advantage plan uses a QHIN to gather patient histories from providers for care management – measuring impact on outcomes. Early lessons could inform changes to TEFCA policy or standards (for instance, ensuring the data payload available through TEFCA includes claims or enrollment data if needed). Another suggestion is aligning TEFCA with CMS's own data exchange programs: if CMS requires MA plans to connect via TEFCA for certain transactions (e.g., when forwarding encounter data to CMS or exchanging information with other payers during beneficiary transitions), that mandate would

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spur adoption.

Digital Identity and Security for Payers: Payers also manage portals for members and providers; supporting common digital identities can improve user experience. We propose that CMS encourage payers to accept patient digital credentials (Login.gov, etc.) for member portal access. A Medicare beneficiary with a Login.gov account, for example, could use it to access their MA plan's site if the plan enabled OIDC federated login. This removes friction for seniors who currently juggle logins. Similarly, for provider-facing portals, a physician's unified digital ID (as discussed earlier) could allow access across multiple plan portals without separate credentials. Security is paramount – these identity solutions must meet multi-factor authentication and other CMS security requirements. Payers will benefit from this approach by offloading identity proofing to trusted third parties and possibly reducing call center issues with account access. We also see value in a nationwide provider directory with FHIR endpoints that payers can use. For payers, having a consistent directory helps with electronic delivery of claims data or prior auth requests to providers. CMS's plans to build such a directory should incorporate payer input on necessary fields (e.g., linking provider endpoint to plan network status or contracting info might be useful for payers to route queries).

Reducing Provider Burden – Payer Contribution: Payers can be partners in reducing the documentation and reporting burden on providers. As mentioned, one idea is for payers to take on some quality measurement tasks. Many payers already calculate HEDIS or other metrics; if CMS facilitates a way for payers to ingest raw clinical data (via bulk FHIR from EHRs) and perform calculations for providers, it could eliminate duplicate efforts. For example, instead of each ACO or practice extracting data for CMS and each MA plan separately, a unified data submission pipeline could feed both CMS and payers. Payers expressed interest in EHR advances enabling bulk extraction of quality data so that plans can do the heavy analytics. We support CMS convening multi-payer pilots where a provider shares one bulk data file that multiple authorized payers can securely access for quality purposes, so the provider doesn't have to respond to numerous quality data requests. Additionally, payers can help providers by sharing timely data that providers may not have, such as notifications of patient hospital admissions or medication fills. CMS could require payers to implement event notification APIs (many already send ADT notifications). This would particularly aid value-based care providers in managing populations.

Finally, payers should also be engaged in information blocking compliance. While payers are considered "actors" under ONC's information blocking regulations, there's less focus on them compared to providers and EHR vendors. CMS should encourage payers to report instances when providers or IT developers block data sharing with plans (for example, a provider refusing to honor the Patient Access API). Greater participation in ONC's complaint process by payers could shed light on systemic issues and reinforce that all actors are accountable.

<u>Recommendations for Payers:</u>

- Enhance API Offerings: Mandate full implementation of FHIR-based Patient Access, Provider Access, and Payer-to-Payer APIs across all CMS-regulated plans. Promote utilization by demonstrating benefits (better care coordination, reduced manual data

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exchange).

- Leverage TEFCA: Address TEFCA limitations by developing payer-specific exchange use cases (risk adjustment, prior auth, etc.) and possibly requiring plans to use TEFCA for certain data submissions. Provide funding or incentives for early adopter payers/QHINs and adjust policies based on pilot outcomes.
- Unified Identity and Directories: Support federated patient identity – e.g. allow Login.gov for plan portals – to simplify beneficiary access. Contribute to building a national provider/FHIR endpoint directory and use it for efficient data exchange with providers.
- Partner to Reduce Burden: Collaborate with providers on quality reporting – accept bulk data from EHRs and perform measure calculations, rather than requiring providers to report separately. Share admission and medication data with providers in real time to aid care management (e.g., via notifications or APIs).
- Promote Data Exchange Compliance: Lead by example in information sharing. Submit information blocking complaints when encountered and ensure the plan's own vendors (like utilization management systems) are not inadvertently blocking data flow. By taking an active role, payers can reinforce a culture of interoperability and signal its importance to their contracted providers.

Technology Vendors, Data Providers, and Networks (TD)

Stimulating a Vibrant Health Tech Ecosystem: A key goal of this RFI is to unleash innovation in digital health products for Medicare beneficiaries. Many health IT startups focus on commercial populations or younger consumers, perceiving Medicare's tech landscape as challenging. CMS can change this by *directly engaging developers* and lowering barriers to entry. We recommend that CMS host regular developer forums and FHIR code-a-thons centered on Medicare use cases (for example, a challenge to build a care navigation app for caregivers of Alzheimer's patients). Prize funds or recognition from CMS could spur interest. CMS might also provide synthetic data and sandboxes that mirror Medicare data (claims, clinical, drug pricing, etc.) so that innovators can build and test solutions easily. *Massive Bio's experience is instructive:* their SYNERGY-AI platform and Clinical Network were built by leveraging diverse data (clinical records, genomic data, trial databases) to create real-time services like patient-trial matching. To attract more companies to build such tools for Medicare, CMS should ensure that its own data (via APIs) is rich and easily accessible (with appropriate privacy controls). For instance, making de-identified Medicare datasets available for AI training could encourage development of AI-driven health apps targeting senior health issues. In the short term, CMS can signal commitment by adding new data to APIs – developers have specifically asked what additional CMS data sources would be most valuable. Many likely desire *integrated claims+clinical data*; while CMS mostly has claims, linking Medicare claims with outcomes from surveys or partnering with NIH for cancer registry data could be powerful. CMS could also consider opening APIs for data such as *durable medical equipment usage, functional status from assessments, or social determinants of health collected by plans*.

Beyond CMS-held data, aligning policy to incorporate *non-traditional data sources* is key. Developers find value in data like device readings, patient-reported outcomes, and genomic information alongside claims. We suggest CMS explore ways to endorse or incorporate such data. For example, CMS could pilot an "App Gallery" where third-party apps that beneficiaries use (e.g., glucometer apps or fitness trackers) can contribute data to a patient's record with consent. This

would require APIs and data standards (perhaps using FHIR Observation resources). By embracing these sources, CMS catalyzes a broader ecosystem where Medicare patients benefit from wellness and disease management apps just as younger populations do.

Digital Identity and Security (Technical Perspectives): From a technology vendor or HIE perspective, implementing digital identity solutions raises practical questions. The consensus is that modern API ecosystems need strong, federated identity for security and convenience. Requiring digital identity credentials (such as NIST IAL2 verified) for data access could enhance cybersecurity by reducing weak passwords and tying access to verified individuals. Using OpenID Connect (OIDC) as the standard protocol for identity federation is advisable – it builds on OAuth2 and is widely adopted, meaning many developers and products already support it. Mandating OIDC across Medicare APIs and networks would ensure compatibility and ease integration. Potential downsides that tech stakeholders note include the overhead of managing these identity services and ensuring equitable access (some patients without smartphones or photo IDs might struggle to obtain digital credentials). To mitigate that, multiple options (government-provided login like Login.gov, as well as private like ID.me or state ID services) should be accepted to give choice. Overall, technology vendors favor clear, uniform standards – we recommend CMS/ONC formally adopt OpenID Connect for any patient or provider-facing authentication in CMS programs. This could be done via rulemaking or guidance that any API requiring login must support OIDC with a set of certified Identity Providers. This approach echoes how SMART on FHIR already relies on OIDC for authentication tokens, and expanding it will strengthen the trust framework.

Open Standards vs. Proprietary APIs: Many EHR vendors and data companies historically offered proprietary APIs, which fragmented integration efforts. CMS should firmly encourage use of open, standards-based APIs. One lever is through certification and payment incentives: CMS could require that, to participate in Medicare (as an ACO, EHR for Meaningful Use, etc.), any data exchange must use standards unless an exception is justified. Another idea is a compliance scorecard published by CMS or ONC that shows which vendors use proprietary interfaces versus standardized FHIR for various functions – essentially public shaming to drive conformity. The benefits of open APIs (lower integration cost, vendor-agnostic apps, easier data aggregation) far outweigh the effort of standardization. For instance, had lab systems universally adopted LOINC and FHIR, the industry would have been better positioned to aggregate COVID-19 data quickly. CMS should coordinate with ONC to expand required API capabilities in certification (e.g., not just patient data APIs but scheduling APIs, or bulk data APIs for all vendors). If a vendor must choose between maintaining a custom API or using FHIR that meets CMS policy, they will lean towards the latter to avoid losing customers or certifications.

Provider Directory and Network Infrastructure: A repeating theme is the need for a unified provider directory. Technologists stress that a centralized directory of FHIR endpoints, identities, and even digital certificates would greatly simplify network connectivity. We support CMS's inquiry into this and suggest that CMS partner with ONC's Recognized Coordinating Entity (Sequoia Project) to build on existing directory efforts (such as those in TEFCA or the NPI database). The directory should be open access or low-cost to maximize use. If funding is an issue, CMS could subsidize it because the public-good nature of such infrastructure is clear – it will help

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patients, providers, payers alike to discover where to send or retrieve data. Additionally, CMS should consider the sustainability of network infrastructure. Operating national networks or directories has costs; policies might be needed to ensure that smaller HIEs or new entrants aren't shut out due to high fees. A possible model is a tiered fee structure or federal support for core interoperability utilities to ensure broad participation.

Another consideration is rationalizing overlapping networks and standards. Many tech stakeholders are concerned about multiple parallel channels (e.g., direct messaging, FHIR point-to-point, various HIE frameworks, and now TEFCA). Redundancy can waste resources and confuse implementers. CMS, with ONC, could convene a summit to map out which standards/channels best fit which use cases and where consolidation might help. For instance, if TEFCA can handle certain queries, perhaps CMS could retire or de-emphasize older mechanisms (like certain direct trust exchanges) to focus industry effort. Conversely, if some innovative protocol emerges (like a secure RESTful push for event notifications), CMS can highlight that as preferred. The end goal should be an "API unification" where a handful of well-defined interoperable services cover most needs, reducing the need to implement dozens of interfaces.

USCDI and Beyond – Data Content Standards: The USCDI provides a core set of data elements, and it has grown over time (now including more social and behavioral data, for example). Tech developers generally appreciate USCDI's role but also report limitations. Many find that USCDI does not yet encompass all data they need – for example, genomic data, advanced imaging, or certain patient-reported outcomes are not fully represented. CMS asked whether adding more elements to USCDI would help or create scope creep. We suggest a balanced approach: incrementally expand USCDI in areas that directly impact care coordination and quality, but accompany expansions with implementation guidance to avoid confusion. High-value additions could include: device data (e.g. blood glucose readings), more granular mental health data, and data relevant to post-acute care. On the other hand, USCDI might not need to include everything if other methods can supply it. The RFI intriguingly mentions if a less structured but more comprehensive format would be preferred given improved language processing. This could hint at using NLP on clinical notes (essentially exchanging free text plus metadata). There is a place for that: not all clinical insight fits into discrete fields. A possible recommendation is for CMS/ONC to pilot a hybrid data exchange: sending core USCDI discrete data *plus* an unstructured summary (like a Continuity of Care Document or even a plain text clinical note) to ensure nothing is lost. With modern AI, the receiver could extract needed info from the text, addressing gaps in discrete data coverage. This aligns with vendor feedback that while structured data is ideal, it's better to receive *something* (even free text) than nothing for a given concept.

Certification and API Compliance: Many technology developers comply with ONC's Health IT Certification Program, which has historically focused on EHR functionality. The RFI asks if certification should evolve to prioritize API-enabled capabilities over specific feature functions. We agree that certification should increasingly test for interoperability performance – i.e., can the system share and receive data effectively – rather than just internal features. For instance, an EHR's ability to allow bulk export of all data (including notes and images) via API is more impactful for system interoperability than whether it has an internal report printing module. Thus, we suggest ONC refocus certification criteria such that any data element a system captures is

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required to be accessible via an API (with appropriate privacy controls). This stems from the 21st Century Cures Act's mandate that certified APIs provide access to all patient data elements permissible by law. ONC could strengthen enforcement of this by actual data export and exchange testing in certification (making vendors demonstrate pulling a full patient record including clinical notes, for example).

Similarly, certification should ensure EHI (Electronic Health Information) export capabilities move toward standardization. As of 2024, EHRs must support an EHI export (which could be a non-standard format). It would benefit the ecosystem if CMS/ONC now move to require that this export be available via a standard API or at least a consistent format (e.g. bulk FHIR files or standardized CSV). This will ease transitions between systems and data sharing for analytics. Another certification angle is assuring data quality and completeness. CMS asks how policy could motivate providers to respond to API requests with the best possible data. One idea: incorporate into CMS's payment or scoring systems a measure of data completeness via APIs (e.g., an ACO could get credit if it provides a broad set of data to a registry or to CMS through automated export). Also, ONC could certify not just EHRs but data registries or other health IT that play roles in interoperability. If registries were certified to accept FHIR and exchange data out to providers, it could streamline quality measurement and research.

Data Exchange Networks and Compliance: Many networks (HIEs, vendor networks, QHINs) exist, and their connectivity varies. The RFI probes how networks use FHIR today, how many endpoints they connect, etc., to understand the landscape. One trend is the shift from older HL7 v2 and IHE-based exchange to FHIR-based query models. CMS should support network efforts to modernize with FHIR. For example, CMS could fund grants for HIEs to add FHIR API layers on top of their networks, making them more accessible to FHIR apps and national exchange. On network scalability, an open question is the tradeoff between point-to-point integrations vs. shared networks. The proliferation of direct FHIR connections (each app connecting to each provider's API) can be hard to manage; conversely, a large network like TEFCA concentrates the connections but can be bureaucratic. A hybrid likely works: encourage point-to-point for simple bilateral data sharing (patient to single provider) and use networks for broad queries (patient to all providers). CMS can help by ensuring policies align: e.g., a provider fulfilling a patient's access request could do so either by enabling a third-party app (point-to-point) or via an Individual Access Services QHIN (network), and both should satisfy compliance. Clear guidance on such equivalence can help avoid redundant implementations.

Regarding information blocking from the tech vendor side, ONC handles certification and potential penalties for health IT developers. But CMS can complement that by requiring transparent reporting of any interfaces or practices that restrict interoperability. Tech vendors reported various forms of information blocking, from exorbitant fees for interfaces to non-responsiveness to integration requests. CMS could, for instance, mandate that certified EHRs annually attest to their connectivity (how many API calls served, how many third-party apps connected, etc.). This data could flag if a vendor is making it unreasonably difficult for outside apps to connect (an indicator of possible blocking). We also support CMS and ONC examining if certain actor categories (perhaps some specialty HIEs or third-party intermediaries) fall through the cracks of current enforcement and need new disincentives.

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Finally, price transparency is an emerging area where tech innovation and data policy intersect. CMS has hospital price transparency and payer cost-sharing data rules, but the data is often hard to use (large, inconsistent files). Tech innovators are trying to build tools to present patients with out-of-pocket cost comparisons, but they need data in standardized, timely formats. We recommend CMS accelerate the transition from static files to APIs for price transparency. For example, require hospitals to publish their machine-readable chargemasters via API, and payers to expose cost estimator tools via API endpoints that third-party apps can call. Responders note current shortcomings: files lack standardized content, vary in format (JSON vs CSV), and aren't updated frequently. CMS should convene standards bodies to define a FHIR-based standard for price data (perhaps an extension of the CARIN IG for Blue Button, but for prices). Workflows that would benefit include integrating cost info into physician EHRs at the point of ordering (so clinicians can discuss cost with patients). If CMS made such price APIs available and mandated their use in Medicare (for example, a requirement that Part D plans expose formulary prices via API), it would encourage industry to follow. Additionally, CMS could foster solution development by highlighting use cases – e.g., encourage startups to make shopping tools for Medicare supplemental plans or procedure pricing, with the knowledge that richer data is coming. Making price transparency functional ultimately empowers patients and can drive competition on cost and quality.

<u>Recommendations for Technology & Data Stakeholders:</u>

- Fuel Innovation with Data: Provide developers with better Medicare data access – richer Blue Button APIs (more data types) and sample data in sandboxes. Use challenges/prizes to focus innovation on Medicare needs (caregiver apps, chronic disease management for seniors, etc.).
- Adopt Federated Identity & OIDC: Standardize on OpenID Connect for authentication across all CMS-related APIs and exchanges. Require use of accredited identity providers for patient and provider login to enhance security.
- Mandate Open APIs: Use policy levers so that all healthcare IT uses publicly documented FHIR (or similarly open) APIs instead of proprietary ones. Expand ONC certification criteria to cover more standardized API capabilities (e.g., scheduling, bulk data) and enforce "no special effort" access to all data elements via API.
- Provider Directory & Network Infrastructure: Establish a nationwide FHIR endpoint directory with support from HHS (ensure it's comprehensive and affordable). Align networks by phasing out duplicative channels and focusing on scalable models (support TEFCA while allowing flexible point-to-point where appropriate).
- Evolve Data Standards: Continuously update USCDI based on stakeholder input to fill gaps but avoid overload. Encourage use of unstructured data exchange (notes, summaries) alongside standardized fields to improve completeness. Invest in NLP tools to make unstructured data usable so this doesn't compromise data-driven care.
- Reform Certification Focus: Shift ONC certification to test real-world interoperability: require evidence of successful data exchange, use of bulk export, and response to all API queries. Define certification for networks or registries that play roles in data aggregation to ensure they support modern standards.
- Transparency and Compliance: Increase reporting on interoperability metrics (API usage,

connectivity) to spot information blocking by vendors. Close gaps by applying information blocking rules or new disincentives to any actor enabling exchange (HIEs, APIs vendors, etc.) who might currently evade accountability. Advance price transparency by moving from static files to APIs, standardizing content, and integrating cost data into care workflows for real-time decision support.

Value-Based Care Organizations (VB)

Digital Health Adoption in Value-Based Care: Accountable Care Organizations (ACOs) and other Alternative Payment Models (APMs) have strong motivation to keep patients healthy and out of the hospital. Digital health tools can be a force-multiplier in these efforts by enabling proactive, continuous care. Yet, many ACOs underutilize tools like remote monitoring, AI risk stratification, or patient engagement apps. To encourage adoption, CMS should incorporate technology use into APM incentive structures. For example, offer a modest shared savings bonus or advanced payment to ACOs that deploy approved digital health management tools to a certain percentage of their high-risk patients. CMS could also relax certain reporting requirements for ACOs that show they are using real-time data tools effectively – essentially trading paperwork for proven tech-enabled management. Current obstacles include upfront costs, uncertain ROI, and integration challenges. CMS's new $15 million initiative with Massive Bio, recognized under the Cancer Moonshot, illustrates a path: by investing in AI-driven trial matching and remote pre-screening hubs, it aims to reach 50,000+ new patients and scale to 250,000 by 2027. This "hub-and-spoke" model, combining AI analytics with decentralized trial participation, is funded because it promises improved outcomes and equity. CMS could create a *Digital Health Adoption Program* granting similar seed funds or implementation support to ACOs for telehealth, RPM, or AI tools that target Medicare's biggest cost drivers (heart failure, diabetes, etc.). The return on investment can be high: as one study showed, a remote monitoring program in a Medicare population reduced hospitalizations by 59%, saving an estimated $12 million in a couple of years. If ACOs see that kind of tangible benefit, they will eagerly adopt these technologies.

Another lever is integrating key tech capabilities into APM requirements or scoring. CMS asks how AI, analytics, care coordination, etc., can be better integrated into APMs. One idea: in ACO quality measures or in the Innovation Center's model requirements, include measures of digital engagement (e.g., percentage of patients on remote monitoring, or having electronic care plans). Alternatively, CMS could require an ACO's care coordination strategy to explicitly address how they use data analytics and patient-facing tech – making it part of the application and evaluation for programs like the Shared Savings Program.

Essential Health IT Capabilities for Value-Based Care: Successful value-based care relies on effective population health management. Certain IT capabilities are now practically required: care planning tools that span settings, event notifications for transitions of care, patient matching across providers, data normalization for combining EHR data, and easy access to claims and cost data. CMS should ensure that these capabilities are either provided or facilitated for ACOs. For example, CMS could extend its Beneficiary Claims Data API (as used in CPC+ or other models) so that any ACO can retrieve near-real-time claims updates on their assigned beneficiaries. This helps identify care outside the network or gaps. Additionally, patient event notifications (like ADT alerts) are so critical that CMS already mandated hospitals share them (CoPs). CMS could verify

that ACOs are indeed receiving these notifications and if not, find out why – perhaps smaller or rural providers lack the HIE connections. Addressing those gaps (maybe via a CMS-sponsored intermediary or requiring participation in a state HIE) would close the loop.

Massive Bio's model underscores some innovative capabilities that could translate to broader value-based care: their use of genomic data and AI to stratify patients for trials is akin to advanced risk stratification in population health. ACOs might similarly use AI to identify patients who would benefit from specific interventions (e.g., who is likely to be hospitalized in next 3 months). CMS can foster this by sharing more data (as above) and by clarifying that using predictive analytics is encouraged and fits under permissible uses of data. Another key capability is patient engagement and empowerment tools – e.g., patient portals or apps that encourage medication adherence, symptom reporting, etc. ACOs may struggle to get patients using them. CMS could partner with patient advocacy groups to create template engagement materials for ACOs to give to patients, highlighting the value of these digital tools in managing their health.

Data Needs for VBC: Data drives value-based success. APM entities need a wide array of data: not just medical claims, but pharmacy, social determinants, behavioral health, and often functional status data. CMS should ask ACOs each year what data they lack. Likely answers include real-time notification of healthcare use (ER visits, urgent care), Medicaid data for their dual-eligibles, and data from outside providers. Efforts like combining Medicare and Medicaid data in the QRS (quality reporting system) are helpful, but more integration is needed. We recommend CMS accelerate projects like the CMS Data at the Point of Care pilot – expanding it so that any value-based provider can query CMS for claims histories of their patients. Also, provide bulk data extracts of *targeted populations* to ACOs (e.g., all dual-eligible patients' Medicare and Medicaid claims in one file).

Interoperability is sometimes harder in VBC because these organizations span multiple providers and EHR systems. This is why *data standardization and easy exchange* is so important. ACOs often set up data warehouses to collect from members; CMS can ease this by continuing to refine standards like FHIR Bulk Data and standard Quality Improvement Core (QI-Core) profiles for common metrics. Standardizing the data that flows in APMs will reduce custom integration costs.

Certification and Standards to Support VBC: The ONC Certification Program today doesn't explicitly address whether an EHR is "good for ACOs," but perhaps it should. The RFI asks if current certification criteria support value-based care delivery and what more is needed. One gap is that certification has focused on individual patient record exchange and specific criteria (like transitions of care docs), but value-based care needs population-level and analytics capabilities. For instance, it would help if certified EHRs were required to have a bulk export feature (for panel management) – which, in fact, the Cures Act API criteria implicitly encourages. Also, data quality and completeness standards in certification would help ACOs: if an EHR is certified, we should trust that it captures key data like problem lists, medications, etc., in structured form that can be exported. Perhaps certification could include a "VBC readiness" test scenario: e.g., demonstrate that a provider can extract all diabetic patients and their last A1c values via the system's API. If not, that's a gap to fix.

Another capability not in certification: care coordination functions such as care planning or tracking referrals. ACOs often deploy separate care management software because their EHRs lack these features in a usable way. ONC could consider adding criteria or optional certification tracks for care management functionality, which would stimulate EHR vendors to improve or at least allow better integration. Additionally, remote monitoring integration is not explicitly tested in certification. We might want future standards to ensure EHRs can consume patient-generated health data easily (perhaps via FHIR Device and Observation data). An ACO that sends patients home with Bluetooth devices should be able to see those readings in the EHR without custom IT work – this should become a baseline expectation.

Reducing Complexity and Preserving Flexibility: Value-based programs involve diverse types of organizations, from small physician-led ACOs to large integrated systems. Tech requirements must be right-sized. CMS has wisely allowed some flexibility (e.g., smaller ACOs may not have the same resources as bigger ones). We should preserve that by perhaps offering multiple pathways to meet a requirement. For example, if an ACO is required to notify patients' primary care docs of hospital admissions, a large system might do it via an integrated EHR notification, whereas a small practice ACO might use a service from their HIE. CMS can set the outcome-oriented requirement (ensure 90% of admissions result in notification) but let the org choose tech solutions.

CMS also references the Physician Fee Schedule 2024 rule allowing CEHRT (Certified EHR Technology) requirements in advanced APMs to be more flexible based on practice area. We support tailoring tech requirements: for instance, an oncology-focused model might need EHRs to handle genomic data, whereas a primary care model might emphasize e-prescribing and referral exchange. CMS should consult with each APM cohort to identify specific needs and then possibly provide a list of acceptable alternative tech solutions. If a small or rural provider cannot implement a full EHR module, maybe a smaller certified app that plugs in could suffice for compliance – this could lower the barrier to entry for resource-constrained participants. We must ensure flexibility doesn't equal lack of interoperability though; any alternative should still adhere to core data standards.

Overcoming Interoperability Challenges in VBC: Many ACOs report issues like difficulty matching patients across different EHRs, lag in data reporting (claims often arrive months later), and incompatibility between systems. A specific challenge is that post-acute facilities (SNFs, home health) and behavioral health providers often are outside the main data streams. CMS can aid by expanding programs like event notification requirements to more settings (as hinted in the RFI about hospital CoP notifications vs. APM participants receiving them). If a patient is in an ACO, any hospital admission notification should reach the ACO care team, not just a community provider. Perhaps CMS could use its authority to require that ACOs (or their participants) be included in the notification routing (this might be achieved via the provider directory concept – an ACO could register as an interested party for certain patients).

Another approach: CMS could stand up a centralized ACO data hub that all ACOs can plug into. For example, Medicare could push claims or utilize a shared FHIR repository where each ACO can see their patients' Medicare utilization in near-real-time. This would mitigate the need for each ACO building custom data pipelines.

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Digital identity can assist here as well – implementing patient digital identity could improve matching patients across datasets (if the same patient uses a common identifier, linking records becomes easier). CMS could pilot using a unique digital ID in an ACO: patients get a credential that they use at all points of care, and the data flows tied to that ID. This is futuristic but worth exploring as a solution to the persistent patient matching problem.

Recommendations for Value-Based Care Organizations:

- Incentivize Tech Investment: Provide bonus payments or shared savings credit for ACOs that deploy proven digital health interventions (RPM, telehealth, AI for risk stratification) that improve outcomes. Offset startup costs via model participation grants.
- Embed Tech in Program Design: Require or reward specific tech-enabled capabilities in APMs – e.g., care coordination platforms, event notifications, patient engagement rates. Ensure each APM has a data exchange plan (with HIEs, etc.) and hold participants accountable for using it.
- Data Access for ACOs: Expand CMS data sharing – give ACOs timely claims feeds, ensure they get hospital ADT notifications (perhaps mandate inclusion of ACO care coordinators in hospital notice workflows). Promote inclusion of SDOH and behavioral health data in ACO datasets to treat the whole patient.
- Align Certification with VBC Needs: Update ONC certification to include population health and care management criteria. Push EHR vendors to support bulk data export, multi-provider care plans, and integration of patient-generated data – features crucial for VBC success.
- Flexible but Interoperable Requirements: Tailor tech requirements to different APM contexts (specialty models vs primary care), and allow smaller providers to use modular tools as long as they meet interoperability standards. Monitor that flexibility does not create data silos.
- Address Key Interop Challenges: Support universal patient identity solutions to improve record matching across ACO participants. Provide tools or services for ACOs to aggregate data (e.g., a CMS-facilitated data hub). Enforce that data blocking has no place in VBC – all participant providers must share data with the ACO and each other, or face consequences (such as removal from the ACO or payment penalties).

Conclusion: In sum, our recommendations aim to build a patient-centric, innovation-friendly health technology ecosystem that empowers patients and caregivers with their data, reduces burden on providers, leverages the strengths of payers, and aligns technology with value-based care goals. By adopting modern APIs (FHIR, SMART on FHIR), strengthening digital identity (OpenID Connect, NIST standards), and fostering public-private collaboration, CMS and ONC can accelerate the transition to an interoperable, digitally enabled healthcare system. Massive Bio's work – from AI trial matching to decentralized trial hubs – exemplifies how data and technology can break down barriers and improve outcomes in real-world practice. We urge CMS to take bold but thoughtful actions: fund what works, regulate where needed to ensure fair data access, and always center policy on improving patient care and health equity. With these steps, the vision of a seamless, secure, and innovative health ecosystem for all Medicare beneficiaries is within reach.

Massive Bio stands ready to collaborate with the government, academia, and other industry partners to advance this critical national priority. We are optimistic that, with the right investments and policies, the United States will remain the world leader in trustworthy AI for health, and in doing so, will secure a future where innovation and compassion go hand in hand.

We thank you for considering our input and for your commitment to harnessing AI for the public good.

Sincerely,

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